No.

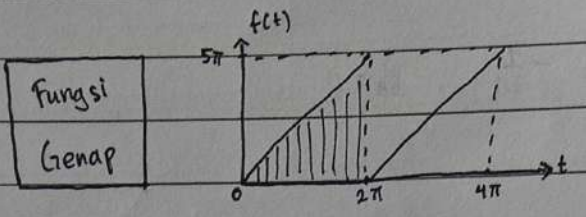
Date:

Latihan

Tentukan deret Fourier fungsi berikut :

1. $f(t) = 2{,}5t$; dengan $0 < t < 2\pi$, periode 2π

$f(0) = 0$, $f(2\pi) = 5\pi$



Fungsi Genap

$\omega = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$

$a_0 = \frac{1}{T}\int_0^T f(t)\,dt$

$a_0 = \frac{1}{2\pi}\int_0^{2\pi} 2{,}5t\,dt$

$= \frac{1}{2\pi} \cdot \frac{2{,}5}{2} t^2 \Big|_0^{2\pi}$

$= \frac{1}{2\pi}\left[\frac{2{,}5}{2}(2\pi)^2 - 0\right]$

$= \frac{1}{2\pi}\left(\frac{2{,}5}{2} \cdot 4\pi^2\right)$

$= \frac{1}{2\pi} \cdot 5\pi^2 = \frac{5}{2}\pi$

$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega t\,dt$

$a_n = \frac{2}{2\pi}\int_0^{2\pi} 2{,}5t \cdot \cos nt\,dt$

$\int at \cdot \cos nt = \frac{at}{n}\sin nt + \frac{a}{n^2}\cos nt$

$\int at \cdot \sin nt = -\frac{at}{n}\cos nt + \frac{a}{n^2}\sin nt$

$= \frac{1}{\pi}\left[\frac{2{,}5t}{n}\sin nt + \frac{2{,}5}{n^2}\cos nt\right]_0^{2\pi}$

$= \frac{2{,}5}{\pi}\left[\frac{t}{n}\sin nt + \frac{1}{n^2}\cos nt\right]_0^{2\pi}$

$= \frac{2{,}5}{\pi}\left[\frac{2\pi}{n}\sin n2\pi + \frac{1}{n^2}\cos n2\pi - \left(\frac{0}{n}\cdot\sin 0 + \frac{1}{n^2}\cos 0\right)\right]$

$= \frac{2{,}5}{\pi}\left[0 + \frac{1}{n^2} - (0 + \frac{1}{n^2})\right]$

$= \frac{2{,}5}{\pi}\left[\frac{1}{n^2} - \frac{1}{n^2}\right] = 0$

$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega t\,dt$

$= \frac{2}{2\pi}\int_0^{2\pi} 2{,}5t\sin nt\,dt$

$= \frac{1}{\pi}\left[-\frac{2{,}5t}{n}\cos nt + \frac{2{,}5}{n^2}\sin nt\right]_0^{2\pi}$

$= \frac{2{,}5}{\pi}\left[\frac{t}{n}\cos nt + \frac{1}{n^2}\sin nt\right]_0^{2\pi}$

$= \frac{2{,}5}{\pi}\left[\frac{2\pi}{n}\cos n2\pi + \frac{1}{n^2}\sin n2\pi - \left(\frac{0}{n}\cos 0 + \frac{1}{n^2}\sin 0\right)\right]$

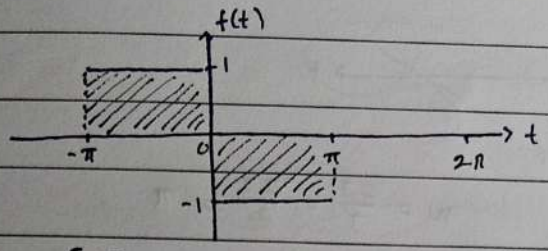
$= \frac{2{,}5}{\pi}\left[\frac{2\pi}{n} + 0 - (0)\right]$

$= \frac{2{,}5}{\pi}\left[\frac{2\pi}{n}\right] = \frac{5}{n}$

Deret Fourier $f(t) = \frac{5}{2\pi} + \frac{5}{1}\sin t + \frac{5}{2}\sin 2t + \frac{5}{3}\sin 3t + \cdots$

2. $f(t) = \begin{cases} 1 & \text{bila } -\pi \le t < 0 \\ -1 & \text{bila } 0 \le t < \pi \end{cases}$ dan $f(t + 2\pi) = f(t)$

$w = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$



[Fungsi Ganjil]

$a_0 = 0$, karena $f(t)$ adalah fungsi ganjil

$a_n = 0$, karena $f(t)$ adalah fungsi ganjil

$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin nwt \, dt$

$b_n = \frac{2}{2\pi}\left[\int_{-\pi}^{0} 1 \sin nt \, dt + \int_{0}^{\pi} -1 \sin nt \, dt\right]$

$= \frac{1}{\pi}\left[-\frac{1}{n} \cos nt \Big|_{-\pi}^{0} + \frac{1}{n} \cos nt \Big|_{0}^{\pi}\right]$

$= \frac{1}{\pi}\left[-\frac{1}{n} \cos 0 - \left(-\frac{1}{n} \cos n\pi\right) + \frac{1}{n} \cos n\pi - \left(\frac{1}{n} \cos 0\right)\right]$

$= \frac{1}{\pi}\left[-\frac{1}{n} + \frac{1}{n} \cos n\pi + \frac{1}{n} \cos n\pi - \frac{1}{n}\right]$

$= \frac{1}{\pi}\left[-\frac{2}{n} + \frac{2}{n} \cos n\pi\right]$

n ganjil $\rightarrow$ $b_n = \frac{1}{\pi}\left[-\frac{2}{n} + \frac{2}{n} \cos \pi\right]$

$= \frac{1}{\pi}\left[-\frac{4}{n}\right] = \frac{-4}{\pi n}$

n genap $\rightarrow$ $b_n = \frac{1}{\pi}\left[-\frac{2}{n} + \frac{2}{n} \cos 2\pi\right]$

$= \frac{1}{\pi}\left[-\frac{2}{n} + \frac{2}{n}\right] = 0$

$b_n \begin{cases} 0 & \text{ketika } n \text{ genap} \\ \frac{-4}{\pi n} & \text{ketika } n \text{ ganjil} \end{cases}$

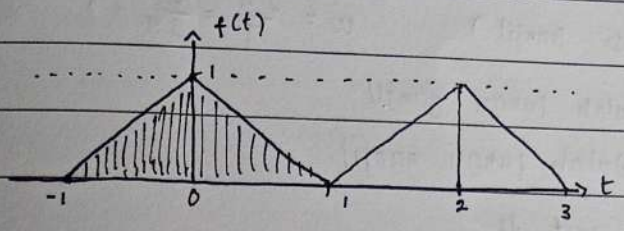
Deret Fourier $f(t) = -\frac{4}{\pi} \sin t - \frac{4}{3\pi} \sin 3t - \frac{4}{5\pi} \sin 5t + \ldots$

3. $f(t) = \begin{cases} t+1 & \text{bila } -1 \le t < 0 \\ -t+1 & \text{bila } 0 \le t < 1 \end{cases}$

$f(0) = 0+1 = 1$, $f(-1) = -1+1 = 0$

$f(0) = 0+1 = 1$, $f(1) = -(1)+1 = 0$

(fungsi genap)



$f(t+2) = f(t)$

$\omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$

$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} f(t)\,dt$

$= \frac{1}{2}\int_{-1}^{0} t+1\,dt + \int_{0}^{1} -t+1\,dt$

$= \frac{1}{2}\left[\frac{t^2}{2} + t \Big|_{-1}^{0} + \left(-\frac{t^2}{2} + t\right)\Big|_{0}^{1} \right]$

$= \frac{1}{2}\left[0 - \left(\frac{1}{2} - 1\right) + -\frac{1}{2} + 1 - 0\right]$

$= \frac{1}{2}\left[\frac{1}{2} + \frac{1}{2}\right] = \frac{1}{2}$

$a_n = \frac{2}{T}\int_{0}^{T/2} f(t)\cos n\omega t\,dt$

$= \frac{2}{2}\left[\int_{-1}^{0} t+1 \cos n\pi t\,dt + \int_{0}^{1} -t+1 \cos n\pi t\,dt\right]$

$= \frac{t+1}{n\pi}\sin n\pi t + \frac{1}{(n\pi)^2}\cos n\pi t \Big|_{-1}^{0} + \frac{-t+1}{n\pi}\sin n\pi t - \frac{1}{(n\pi)^2}\cos n\pi t \Big|_{0}^{1}$

$= \frac{1}{n\pi}\sin 0 + \frac{1}{(n\pi)^2}\cos 0 - \left(\frac{0}{n\pi}\sin(-n\pi) + \frac{1}{(n\pi)^2}\cos(-n\pi)\right) +$

$\frac{0}{n\pi}\sin n\pi - \frac{1}{(n\pi)^2}\cos n\pi - \left(\frac{1}{n\pi}\sin 0 - \frac{1}{(n\pi)^2}\cos 0\right)$

$= \frac{1}{(n\pi)^2} - \frac{1}{(n\pi)^2}\cos n\pi - \frac{1}{(n\pi)^2}\cos n\pi + \frac{1}{(n\pi)^2}$

$= \frac{2}{(n\pi)^2} - \frac{2}{(n\pi)^2}\cos n\pi$

n genap → $a_n = \frac{2}{(n\pi)^2} - \frac{2}{(n\pi)^2}\cos 2\pi = 0$

n ganjil → $a_n = \frac{2}{(n\pi)^2} - \frac{2}{(n\pi)^2}\cos \pi = \frac{4}{(n\pi)^2}$

$a_n = \begin{cases} 0 & \text{ketika n genap} \\ \frac{4}{(n\pi)^2} & \text{ketika n ganjil} \end{cases}$

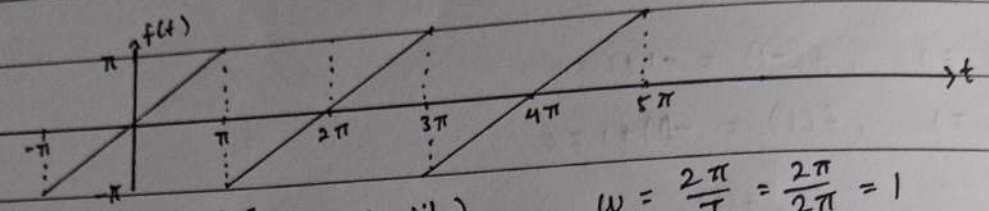
$b_n = 0$, karena $f(t)$ adalah fungsi genap dan sin adalah fungsi ganjil, maka ketika

fungsi genap × fungsi ganjil = fungsi ganjil $= 0$

Deret Fourier $= \frac{1}{2} + \frac{4}{\pi^2}\cos \pi t + \frac{4}{9\pi^2}\cos 3\pi t + \frac{4}{25\pi^2}\cos 5\pi t + \frac{4}{49\pi^2}\cos 7\pi t + \ldots$

Date:

4. $f(t) = t$ bila $-\pi < t < \pi$, dan $f(t + 2\pi) = f(t)$, lihat gambar berikut:



(Fungsi Ganjil) $\omega = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$

$a_0 = 0$, karena $f(t)$ adalah fungsi ganjil

$a_n = 0$, karena $f(t)$ adalah fungsi ganjil

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin n\omega t \, dt$$

$$= \frac{2}{2\pi} \int_{-\pi}^{\pi} t \sin nt \, dt$$

$$= \frac{1}{\pi} \left[-\frac{t}{n} \cos nt + \frac{1}{n^2} \sin nt \right] \Big|_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \left(\frac{\pi}{n} \cos(-n\pi) - \frac{1}{n^2} \sin n\pi \right) \right.$$

$$= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi - \frac{\pi}{n} \cos n\pi \right]$$

$$= \frac{1}{\pi} \left[-\frac{2\pi}{n} \cos n\pi \right]$$

$$= -\frac{2}{n} \cos n\pi$$

n genap $\rightarrow$ $b_n = -\frac{2}{n} \cos 2\pi = -\frac{2}{n}$

n ganjil $\rightarrow$ $b_n = -\frac{2}{n} \cos \pi = \frac{2}{n}$

Deret Fourier $f(t) = \frac{2}{1} \sin t - \frac{2}{2} \sin 2t + \frac{2}{3} \sin 3t - \frac{2}{4} \sin 4t + \ldots$